c) capacity to generate hybrid protoplasts.
d) recovery of healthy plants from diseased plants.

2. Micro propagation involves
   a) vegetative multiplication of plants by using micro-organisms.
   b) vegetative multiplication of plants by using small explants.
   c) vegetative multiplication of plants by using microspores.
   d) Non-vegetative multiplication of plants by using microspores and megaspores.

3. Match the following :

Column A	Column B
1) Totipotency	A) Reversion of mature cells into meristerm
2) Dedifferentiation	B) Biochemical and structural changes of cells
3) Explant	C) Properties of living cells develops into entire plant
4) Differentiation	D) Selected plant tissue transferred to culture medium

	1	2	3	4
a)	C	A	D	B
b)	A	C	B	D
c)	B	A	D	C
d)	D	B	C	A

4. The time duration for sterilization process by using autoclave is _____ minutes and the temperature is _____
   a) 10 to 30 minutes and 125° C
   b) 15 to 30 minutes and 121° C
   c) 15 to 20 minutes and 125° C
   d) 10 to 20 minutes and 121° C

5. Which of the following statement is correct
   a) Agar is not extracted from marine algae such as seaweeds.
   b) Callus undergoes differentiation and produces somatic embryoids.
   c) Surface sterilization of explants is done by using mercuric bromide
   d) P^H of the culture medium is 5.0 to 6.0

6. Select the incorrect statement from given statement
   a) A tonic used for cardiac arrest is obtained from Digitalis purpuria
   b) Medicine used to treat Rheumatic pain is extracted from Capsicum annum
   c) An anti malarial drug is isolated from Cinchona officinalis.
   d) Anti-cancinogenic property is not seen in Catharanthus roseus.

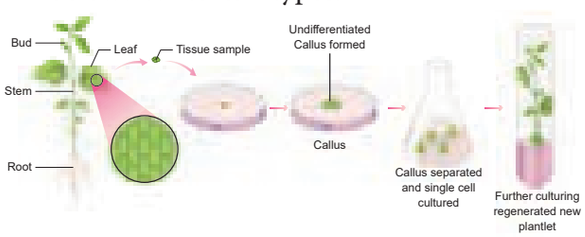
7. Virus free plants are developed from
   a) Organ culture
   b) Meristem culture
   c) Protoplast culture
   d) Cell suspension culture

8. The prevention of large scale loss of biological interity
   a) Biopatent b) Bioethics
   c) Biosafety d) Biofuel

9. Cryopreservation means it is a process to preserve plant cells, tissues or organs
   a) at very low temperature by using ether.
   b) at very high temperature by using liquid nitrogen
   c) at very low temperature of -196 by using liquid nitrogen
   d) at very low temperature by using liquid nitrogen

10. Solidifying agent used in plant tissue culture is
    a) Nicotinic acid b) Cobaltous chloride
    c) EDTA d) Agar

11. What is the name of the process given below? Write its 4 types.



12. How will you avoid the growing of microbes in nutrient medium during culture process? What are the techniques used to remove the

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